Emotion-Aware Smart Home

منزل ذكي متفاعل مع المشاعر

Name	ID	Program
Lenda Alaa Deif	806512779	CS
Haneen Ahmed Elbogadady	808967642	AI
Sohila Mohamed Abdelbaky	808967640	AI

Project Idea

Our project presents an emotion-aware smart home prototype that combines real-time facial emotion recognition with Raspberry Pi-based hardware control. A custom CNN trained on a balanced FER dataset classifies the user's facial expression into five emotions: Angry, Happy, Neutral, Sad, and Surprise.

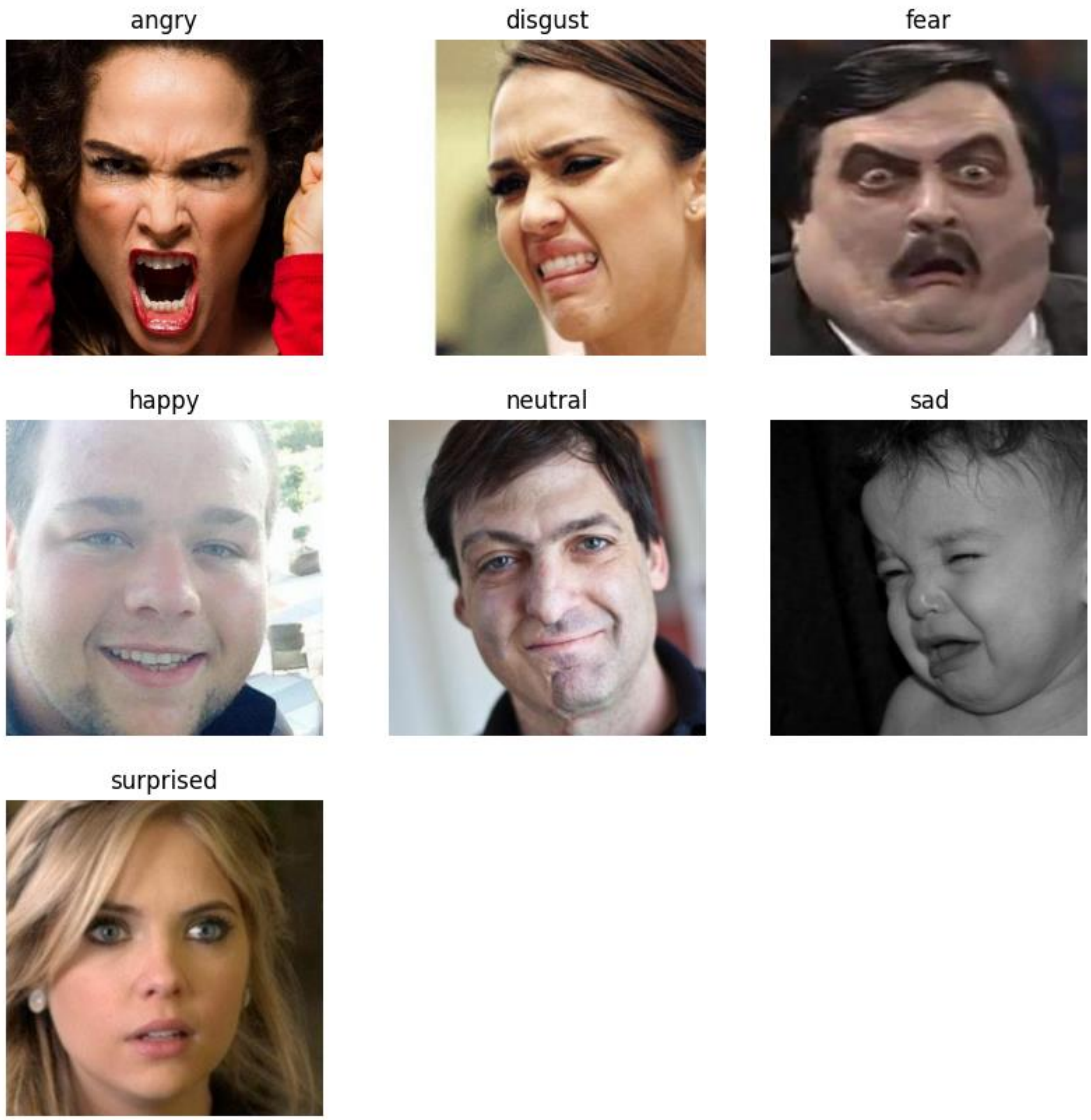
Based on the detected mood, the Raspberry Pi controls physical smart-home outputs that were successfully implemented in the prototype, including an LED strip and a fan. The system adjusts LED brightness and fan speed in real time to create a more adaptive and comfortable environment.

In addition to emotion recognition, the project also includes a sleep detection module based on MediaPipe facial landmarks, Eye Aspect Ratio (EAR), and head pose estimation. This enables the system to respond not only to emotional state, but also to drowsiness and sleep-related conditions

Dataset link:https://universe.roboflow.com/project-group-13/facial_emotion_detection-1vnia/dataset/4

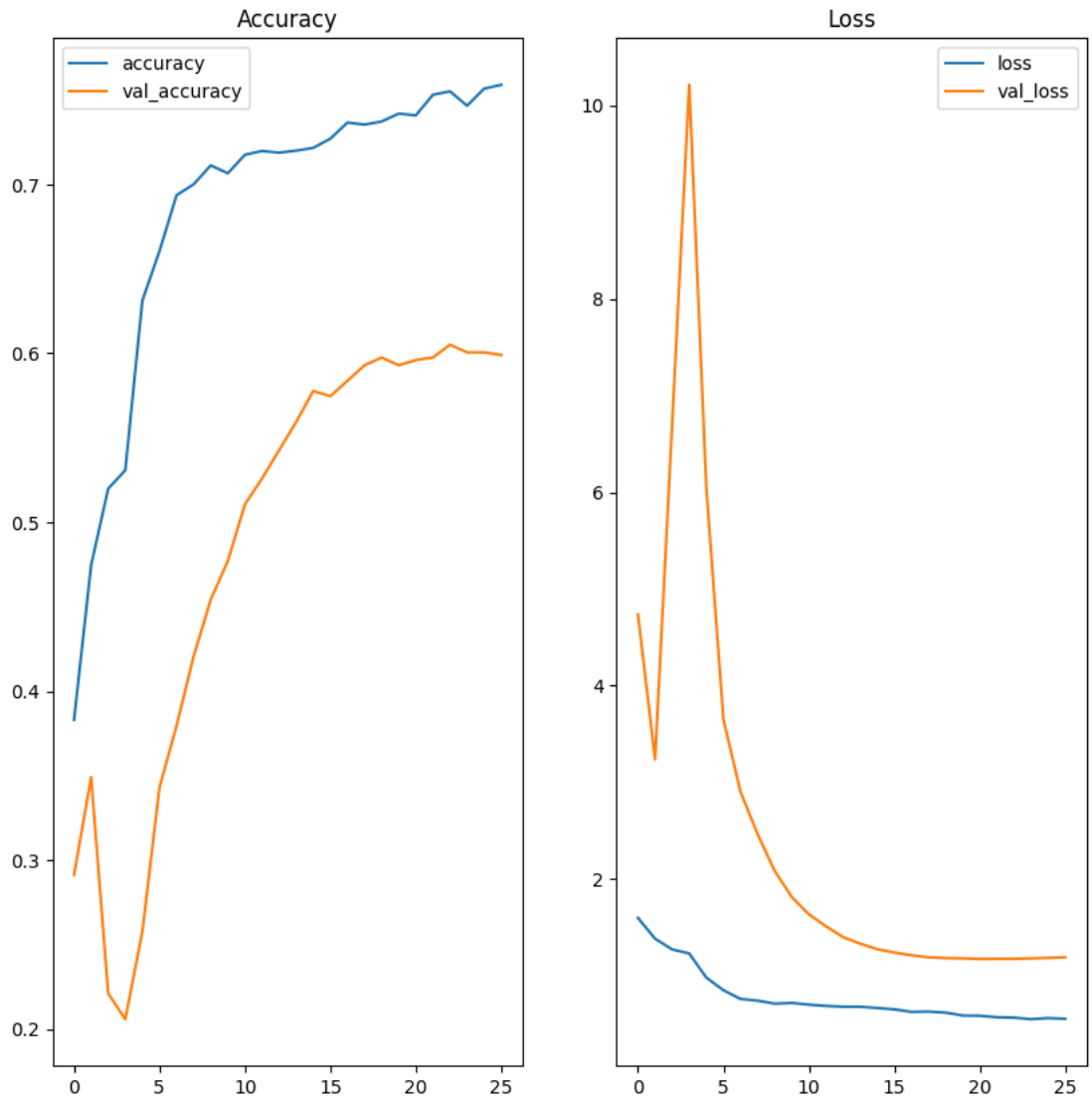
Brief about the data set :our dataset is a Roboflow object detection dataset for facial emotion analysis. It contains 8,695 images in version 4. All images are in the training set, with no validation or test split. It also has no preprocessing or augmentation applied.

Sample:



"The model's performance was limited due to significant visual overlap between classes, reducing its ability to learn discriminative features."

Despite fine-tuning a pre-trained MobileNetV2 model, performance remained limited due to the high similarity between classes.



The obtained results were unsatisfactory in real-time applications, primarily because the model was not optimized for low-resource environments. A lightweight architecture is needed for effective deployment on a Raspberry Pi

"Following several trials and refinements, these results were achieved and considered the most effective."

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Epoch 62/80
183/183 ━━━━━━━━━━━━━━━━━━━━ 0s 439ms/step - accuracy: 0.7318 - f1_score: 0.7076 - loss: 0.8313 - precision: 0.8167 - recall: 0.6242
Epoch 62: val_f1_score did not improve from 0.54192
183/183 ━━━━━━━━━━━━━━━━━━━━ 85s 463ms/step - accuracy: 0.7318 - f1_score: 0.7076 - loss: 0.8313 - precision: 0.8168 - recall: 0.6242 - val_accuracy: 0.5518 - val_f
1_score: 0.5418 - val_loss: 1.3516 - val_precision: 0.6344 - val_recall: 0.4728 - learning_rate: 1.0000e-05
Epoch 63/80
183/183 ━━━━━━━━━━━━━━━━━━━━ 0s 439ms/step - accuracy: 0.7342 - f1_score: 0.7010 - loss: 0.8335 - precision: 0.8148 - recall: 0.6151
Epoch 63: val_f1_score did not improve from 0.54192
183/183 ━━━━━━━━━━━━━━━━━━━━ 85s 462ms/step - accuracy: 0.7341 - f1_score: 0.7010 - loss: 0.8335 - precision: 0.8148 - recall: 0.6151 - val_accuracy: 0.5529 - val_f
1_score: 0.5404 - val_loss: 1.3504 - val_precision: 0.6325 - val_recall: 0.4717 - learning_rate: 1.0000e-05
Epoch 63: early stopping
Restoring model weights from the end of the best epoch: 53.
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Sleep Detection Module

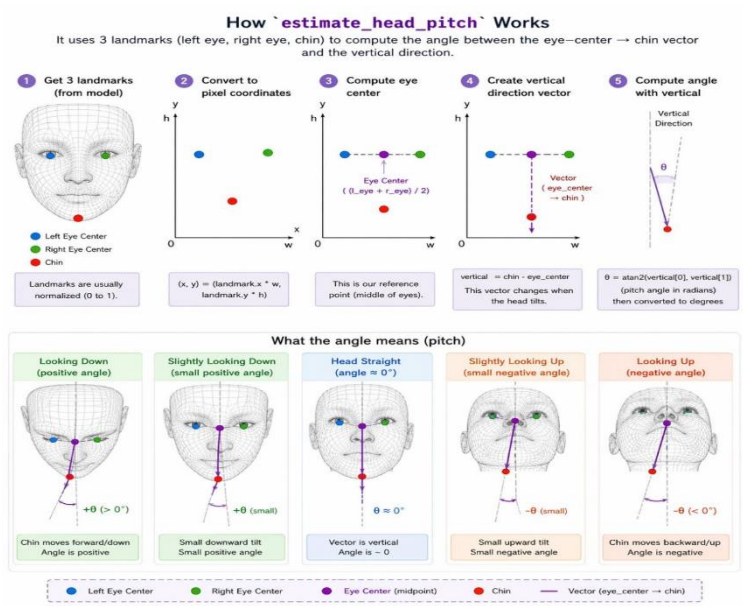
To further enhance system intelligence, we developed an additional sleep detection module. This component leverages the MediaPipe library to extract facial landmarks and analyze user state in real time.

The module focuses on the following key features:

- **Eye Aspect Ratio (EAR):**
  Eye landmarks are used to compute EAR, allowing the system to distinguish between normal blinking, drowsiness, and actual sleep. This ensures that lighting adjustments are based on accurate user state detection rather than transient eye closures.

- **Head Pose Estimation (Pitch Angle):**
  By analyzing the spatial relationship between the eye center and chin landmarks, the system estimates head pitch. This helps detect nodding behavior, which is a strong indicator of drowsiness or sleep.

In addition, face detection is performed using the Haar Cascade Classifier, ensuring efficient and real-time processing when integrated with the landmark-based analysis.



Potential Improvements:

- **Utilizing a more reliable and higher-quality dataset** to improve model generalization and robustness.
- **Deploying on a more powerful Raspberry Pi or edge device**, enabling the use of more complex and accurate models while maintaining real-time performance.